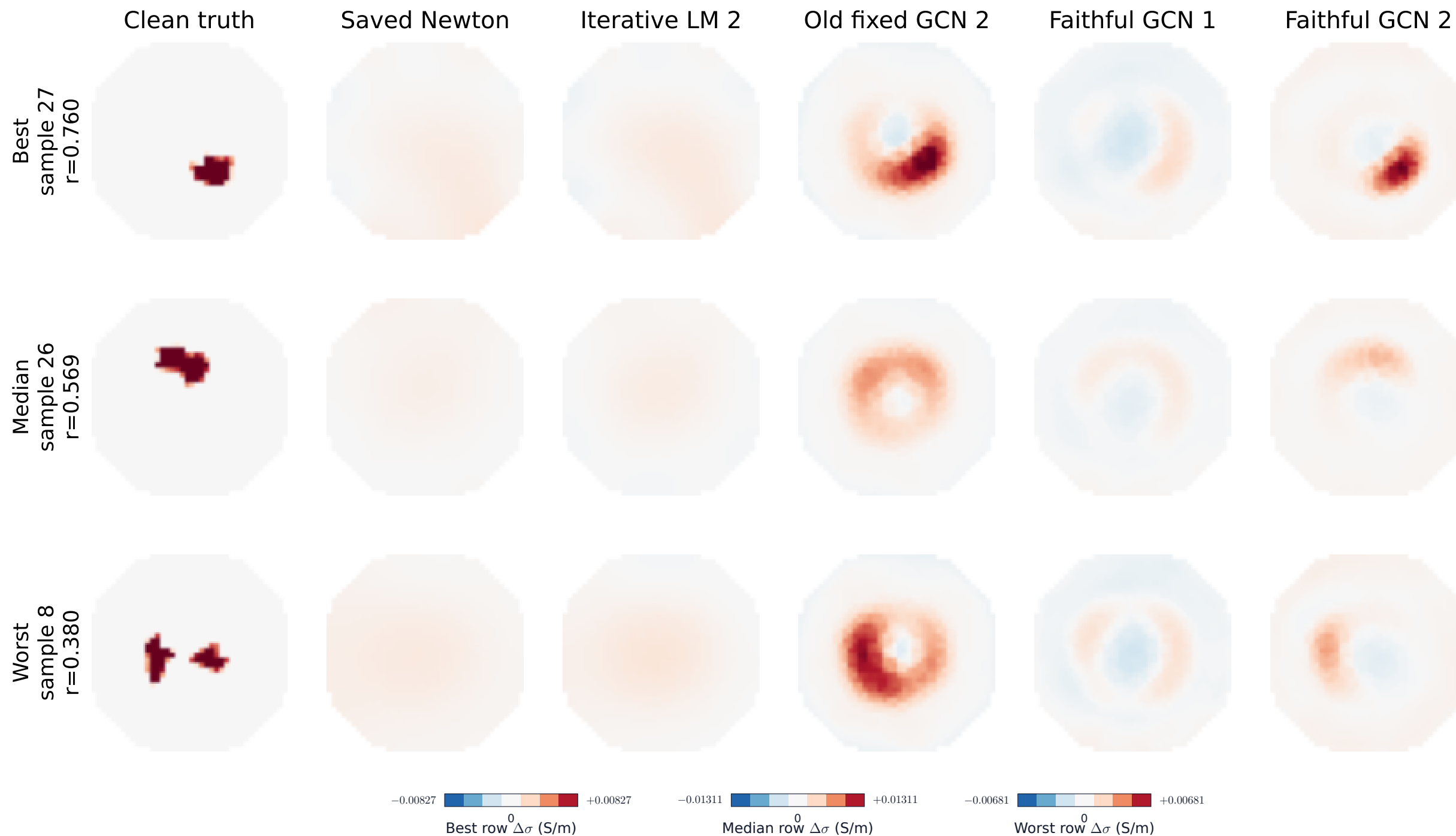


# Selected GCNM: best, median, and worst synthetic cases

Synthetic evaluation – known ground truth  $\Delta\sigma^*$ ; rows are selected objectively by final-stage image correlation

Ground truth and reconstruction evolution: faithful\_background025\_hom\_seed0



**Takeaway.** Objectively selected best, median, and worst cases for the faithful  $\alpha = 1$ ,  $\beta = 0.25$  model. The faithful stages concentrate energy nearer the true vessel region, but still predict a smooth learned distribution rather than resolving every compact vessel—an honest limit with only 32 measurements.

**Purpose.** Directly compares clean truth, the saved Newton step, the two-step physics-only control, the earlier fixed-feature GCN, and the recomputed-physics GCNM across easy, typical, and difficult cases.